

The Effect of Climbing Mount Everest on Spleen Contraction and Increase in Hemoglobin Concentration During Breath Holding and Exercise

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Abstract

Engan, Harald K, Angelica Lodin-Sundstrom, Fanny Schagatay, and Erika Schagatay. The effect of climbing Mount Everest on spleen contraction and increase in hemoglobin concentration during breath holding and exercise. *High Alt Med Biol.* 15:52–57, 2014.—Release of stored red blood cells resulting from spleen contraction improves human performance in various hypoxic situations. This study determined spleen volume resulting from two contraction-evoking stimuli: breath holding and exercise before and after altitude acclimatization during a Mount Everest ascent (8848 m). Eight climbers performed the following protocol before and after the climb: 5 min ambient air respiration at 1370 m during rest, 20 min oxygen respiration, 20 min ambient air respiration at 1370 m, three maximal-effort breath holds spaced by 2 min, 10 min ambient air respiration, 5 min of cycling at 100 W, and finally 10 min ambient air respiration. We measured spleen volume by ultrasound and capillary hemoglobin (HB) concentration after each exposure, and heart rate (HR) and arterial oxygen saturation (SaO₂) continuously. Mean (SD) baseline spleen volume was unchanged at 213 (101) mL before and 206 (52) mL after the climb. Before the climb, spleen volume was reduced to 184 (83) mL after three breath holds, and after the climb three breath holds resulted in a spleen volume of 132 (26) mL ($p=0.032$). After exercise, the preclimb spleen volume was 186 (89) mL vs. 112 (389) mL after the climb ($p=0.003$). Breath hold duration and cardiovascular responses were unchanged after the climb. We concluded that spleen contraction may be enhanced by altitude acclimatization, probably reflecting both the acclimatization to chronic hypoxic exposure and acute hypoxia during physical work.

Key Words: acclimatization, apnea, hematology, hypobaric hypoxia, red cell volume

Introduction

THE RELEASE OF STORED RED BLOOD CELLS resulting from spleen contraction occurs during breath holding (BH) (Bakovic and others, 2003; Schagatay and others, 2001; 2005), simulated altitude by eupneic hypoxia (Richardson and others, 2008), and maximal exercise (Laub and others, 1993). The spleen is among the most densely sympathetically innervated organs (Felten, 2000) and contraction of the human spleen is likely triggered by an sympathoadrenergic mechanism (Schagatay and others, 2001). Both hypoxia and hypercapnia have been shown to trigger spleen contraction (Richardson and others 2009; 2012). The average human

spleen contains a reservoir of approximately 200–250 mL of blood with a splenic hematocrit of about 80%, about twice the hematocrit circulating in blood in the rest of the body (Koga, 1979). Contraction typically increases the total amount of circulating red blood cells between 2%–4% with some individuals responding with of up to 10% increases (Richardson and others, 2008). This response prolongs breath holding duration across breath hold series (Bakovic and others, 2003; Schagatay and others, 2001) and the expulsion of stored red blood cells may temporarily increase the circulating gas storage and CO₂ buffering capacity, improving performance in both normoxic and hypoxic situations in humans. Richardson and others (2008) showed that, in addition to overall

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polycythemia, the transient elevation in Hb concentration during breath holding was attenuated during ascent to higher altitudes, possibly indicating a tonic contraction of the spleen. During descent, the Hb concentration elevation was greater than at similar altitude during ascent (Richardson, 2008).

Few studies have addressed the role of long-term hypoxic exposure on spleen volume. Sonmez and colleagues (2007) found concomitant reductions in spleen volume and increases in hemoglobin (Hb) and hematocrit (Hct) after 3–6 months of chronic exposure to altitude among relocated lowlanders. Larger spleen volumes have been observed in elite freedivers, who as a group are often exposed to hypoxia (Schagatay and others, 2012). In contrast, Engan and colleagues (2013) found no changes in baseline spleen volume or spleen volume reduction during breath holds after 2 weeks of daily breath holding training. Yet augmented spleen contraction in breath holding-trained individuals has previously been reported (Bakovic and others, 2003; Prommer and others, 2007) and breath-hold divers had greater increases in Hb concentration during breath holding compared to elite skiers and untrained individuals (Richardson and others, 2005). This suggests that previous reports of enhanced effects in divers are either due to predisposition or that there may be a training effect, but that more time may be needed for any changes to occur than the few weeks previously studied.

An increase in O₂-carrying capacity resulting from polycythemia plays an important role in acclimatization to high altitude (West, 2012). While it has been shown that spleen contraction occurs as a response to normobaric hypoxia (Lodin and Schagatay, 2010), this response after long-term hypoxic exposure has not previously been investigated. The present study aimed to determine to what extent spleen volume and the contractile function of the spleen were affected by long-term high altitude exposure in climbers ascending Mount Everest, 8848 m above sea level (a.s.l.).

Methods

Subjects

Ten subjects (9 men and 1 woman) preparing for an ascent of Mount Everest volunteered to the study. All subjects received detailed information about the testing procedures and provided informed consent. The study protocol was approved and conducted in accordance with ethical standards of the Regional Committee for Medical and Health Research Ethics in Umeå, Sweden, and Nepal Health Research Council (NHRC), Nepal, and complied with the 2004 Declaration of Helsinki. One of the male western expedition members suffered from acute mountain sickness during ascent and withdrew from the expedition, together with one of the male Sherpas. Both these subjects were subsequently excluded from the study.

Five of the included subjects were Sherpas with a highland origin but current residents of Kathmandu, and three were Caucasian lowlanders. All Sherpas and two lowlanders had summited 8000 m peaks before, and one lowlander had climbed above 7000 m a.s.l. During the month preceding the expedition, two Sherpas had climbed to 3500 m and 5400 m a.s.l., respectively, after which they resumed to Kathmandu 3 weeks prior to the first test. One lowlander had been sleeping occasionally in a normobaric hypoxic tent simulating altitude up to 6000 m a.s.l. before travelling to Nepal, while none of

the remaining subjects had been exposed to altitude conditions during the last month.

Expedition

The expedition left Kathmandu, Nepal, the day after the pre-test, heading for the Tibetan side of Mount Everest by car. They climbed to an altitude of 8848 m a.s.l. via Everest Base Camp (5400 m a.s.l.) and North Col (7002 m a.s.l.) and successfully reached the summit of Mount Everest after 45 days of which they stayed above Everest Base Camp around 35 days. After reaching the summit of Mount Everest, subjects went down to Advanced Base Camp (6340 m a.s.l.) with one overnight before continuing to Everest Base Camp where the expedition spent 2 nights. After an additional 2 nights, the subjects arrived in Kathmandu. The lowlanders followed an individually customized ascending plan based on self-perceived health situation and weather conditions. This included shorter sojourns (from hours to days) at progressively higher altitudes before descending to a lower altitude for a period of time, in a stepwise progression towards the summit to allow sufficient adaptation and recovery. The Sherpa, carrying most of the necessary food and equipment, traversed many of the routes several times in order to organize and supply the different camps.

Experimental protocol

A pre-post intervention study design was utilized with subjects serving as their own control. Pre- and post-expedition tests were identical and conducted in the same laboratory setting in Kathmandu at 1370 m a.s.l. The pre-expedition test was conducted within 2 days prior departure for Tibet, when the lowlanders had then been in Kathmandu 2–4 days. The post-expedition test was conducted within 6–7 days after summiting Mount Everest and after one night sleep in Kathmandu. Subjects arrived at the laboratory after at least 1 h without eating, drinking, or physical exercise. Anthropometric variables were collected (Table 1), and previous and recent history of altitude exposure was noted. The tests consisted of an initial ~5 min period in the sitting position during which calibration of the ultrasound image of the spleen occurred,

TABLE 1. ANTHROPOMETRIC DATA AND BASELINE VALUES

Variables	Pre-altitude	Post-altitude
Age (years)	35 (7)	
Height (m)	1.7 (0.6)	
Body mass (kg)	70.6 (12.7)	64.5 (11.5) $p=0.0048^*$
Waist circumference (cm)	86.6 (9.9)	82.3 (10.6) $p=0.0004^*$
VC (L)		5.3 (1.7)
MAP (mmHg)	95.6 (12.7)	93.7 (11.2) $p=0.6276$
HR (bpm)	74.2 (9.5)	72.7 (13.6) $p=0.7553$
SaO ₂ (%)	96.3 (0.7)	95.7 (0.9) $p=0.2602$
RR (breaths/min)	19.4 (4.0)	21.1 (4.2) $p=0.3682$
ETCO ₂ (%)	5.0 (0.5)	4.2 (0.6) $p=0.0040^*$
Spleen volume (mL)	213.3 (101.0)	205.5 (52.3) $p=0.7925$
Hb (g/L)	144.0 (9.8)	177.1 (17.1) $p=0.0003^*$

Mean (SD) values for all 8 subjects. ETCO₂, end tidal CO₂; Hb, hemoglobin concentration; HR, heart rate; MAP, mean arterial pressure; RR, respiration ratio; SaO₂, arterial oxygen saturation; VC, vital capacity. VC was measured only in the post-test. * indicates significant p value.

followed by the spleen being measured repeatedly (Esaote, MyLab25Gold, Firenze, Italia) for another 5 min. After this, the sitting subjects respired oxygen via a mask at 1 L min^{-1} for 20 min, followed by an additional period of 20 min ambient air respiration. This was followed by three maximal effort breath holds (BH) in ambient air, each spaced by a 2 min breathing period. The duration of intervals between apneas was fixed to 2 minutes because this protocol has been shown to maximize spleen contraction yet provide sufficient time for recovery to normalize in situ O_2 and CO_2 levels between apneas (Schagatay and others, 2005). Subjects were breathing oxygen to avoid any effects of hypoxia on the spleen contraction because the lab was located at 1370 m a.s.l resulting in reduced ambient pressure of oxygen compared to sea level. After oxygen breathing, an additional 20 min of rest breathing ambient air was implemented to allow sufficient time to establish a resting condition from where the apneas and the exercise was initiated, and to avoid breath holding directly after oxygen breathing that would prolong the development of a hypoxic stimulus studied.

Breath holds were performed in the sitting position preceded by spontaneous respiration, and started after a normal tidal volume inspiration (at the peak of the inspiratory phase of the normal breathing cycle) with the request to take a deep but not maximal breath hold at inspiration. After the apneic episodes, the subject rested while breathing ambient air for 10 min, after which the subject stood up and the chair was removed and replaced by a cycle ergometer pre-adjusted for subject's height (Sunburner, USA). The subject cycled for 5 min at 55–60 rpm with a 100 W load after which he/she rested in a sitting position for an additional 10 min.

Measurements

Body mass (kg) and waist circumference (cm) were measured. Vital capacity (L) was measured in triplicate in standing subjects with a spirometer (Microlab Spirometer, Micro Medical Ltd., Kent, UK). After 5 min sitting, blood pressure (mmHg) was measured in duplicate with an automated blood pressure monitor (Omron M41, Omron Healthcare, Europe) on the upper right arm. Heart rate (HR), arterial oxygen saturation (SaO_2), end tidal carbon dioxide (ETCO_2), and breathing frequency (RR) were measured continuously using a combined pulse oxymeter–capnograph (Medair Lifesense LS1-9R, Nonin Medical Inc, Medair AB, Delsbo, Sweden) with data stored via memory unit (TrendSense, Nonin Medical Inc, Medair AB, Hudiksvall, Sweden). Spleen volume was measured via ultrasound every minute throughout the protocol, increasing in frequency to every 30 sec during the 3 min following each change between the different interventions of the protocol to detect any rapid changes. Finger capillary blood samples were collected in microtainer cuvettes in duplicate on five occasions: after 20 min of oxygen respiration, after 20 min of ambient air respiration, following breath hold 3, after cycling and following the final 10 min resting period, and analyzed via a portable hemoglobin analyzer (Hemocue AB, Ängelholm, Sweden).

Data analysis

All data are reported as mean value $\pm$ standard deviation (SD) for the group. Data was analyzed with GraphPad Prism version 5.04 for Windows, (GraphPad Software, La Jolla California USA, www.graphpad.com). Baseline values used for calculation of changes of the continuous variables HR, SaO_2 , ETCO_2 , RR, and spleen volume were obtained from the

5 min period before the oxygen breathing. Spleen volume was calculated according to the Pilström equation: $L\pi(WT-T^2)/3$, where spleen length (L), thickness (T), and width (W) are provided by ultrasound measurement (Schagatay and others, 2005). Reference values for Hb concentration were obtained from the first blood samples taken after 20 min air respiration. Spleen volume changes arising from breath holds, exercise and recovery were calculated by comparing baseline values with those values obtained directly after breath holds, during the 2 last minutes of the exercise period and during the whole recovery period, respectively. Changes in cardiovascular and respiratory variables arising from breath hold and exercise were obtained by comparing baseline values with mean values for the 1-minute period following breath hold termination and for the whole cycling exercise period. During some breath holds, the signal for HR was lost for some subjects, and these recordings were excluded from the analysis. Changes in Hb were obtained by comparing baseline values with mean values obtained immediately after the third breath hold, after exercise, and after recovery.

Differences in baseline subject characteristics between pre- and post-measurements were evaluated with paired Students *t*-tests. Statistical significance was accepted at $p < 0.05$. Non-significant trends were denoted for $p < 0.1$. Changes in spleen size, Hb concentration, and breath hold duration with condition-within each time group (pre-altitude and post-altitude) were assessed using one-way repeated measures (RM) Analysis of Variance (ANOVA). Where appropriate for spleen size and Hb concentration, Bonferroni post hoc analyses were used to determine if condition (breath holds exercise and recovery) differed from baseline. Next, significant differences in condition between pre- and post-altitude measurements were tested for by consideration of the interaction term (time X condition) derived from the two-way RM ANOVA analysis. Where appropriate post hoc testing was done by using a paired Student's *t*-test. *P* values of the post hoc comparisons were corrected according to Bonferroni inequalities.

Results

Body composition

Body mass and waist circumference were reduced by 8.4 (11.5) % and 5.1 (10.5) % across the expedition, respectively (Table 1).

Breath hold duration

Breath hold duration did not differ between pre- and post-test during the series of breath holds (Table 2).

Breath hold duration across test series (BH1, BH2, and BH3) was not significantly increased during pretest, but tended to increase during post-test (Table 2).

Cardiovascular variables

Baseline values for MAP were not different between tests (Table 1). HR during exercise was not different ($p = 0.29$) during pre- and post-tests, at 89.4 (11.7) beats per minute (bpm) and 85.3 (4.55) bpm, respectively.

Respiratory variables and arterial oxygen saturation

Baseline values for SaO_2 and RR did not differ between pre- and post-tests, while the baseline value for ETCO_2 was 16.3 (10.7) % less during post-test (Table 1). Mean (SD) SaO_2

TABLE 2. SUMMARY OF ONE-WAY (CONDITION-EFFECT) AND TWO-WAY (CONDITION, TIME- AND INTERACTION EFFECTS) REPEATED MEASURE ANOVA RESULTS

Test variable	Time	Repeated Measurements (Mean $\pm$ SD)						Analysis of Variance (p Values)			
		Condition 1 Baseline	Condition 2 BH1	Condition 3 BH2	Condition 4 BH3	Condition 5 Exercise	Condition 6 Recovery	1 W-RM W/in group (Condition Effect)	2 W-RM W/subj (Condition Effect)	Time Effect	Interaction
Breath hold duration (sec)	Pre-altitude	—	44.5 $\pm$ 10.1	55.5 $\pm$ 25.7	58.3 $\pm$ 30.2	—	—	0.179	0.005	0.516	0.667
	Post-altitude	—	54.3 $\pm$ 29.1	62.1 $\pm$ 39.9	73.1 $\pm$ 48.7	—	—	0.054			
Spleen volume (ml)	Pre-altitude	213.8 $\pm$ 101.0	192.6 $\pm$ 89.6	185.1 $\pm$ 89.9	175.5 $\pm$ 80.0	186.3 $\pm$ 88.7	194.5 $\pm$ 80.2	0.562	0.100	0.032	0.217
	Post-altitude	205.8 $\pm$ 52.3	126.5 $\pm$ 32.3 $p=0.027$	148.2 $\pm$ 40.4 $p=0.030$	121.6 $\pm$ 36.5 $p=0.010$	111.6 $\pm$ 38.6 $p=0.003$	156.1 $\pm$ 34.8 $p=0.045$	<0.001			
Hemoglobin concentration (g/l)	Pre-altitude	144.0 $\pm$ 9.8	—	—	148.6 $\pm$ 8.3	149.3 $\pm$ 6.9 $p=0.095$	145.4 $\pm$ 9.8	0.133	0.080	<0.001	0.667
	Post-altitude	177.2 $\pm$ 17.4	—	—	182.7 $\pm$ 14.0	183.8 $\pm$ 14.1 $p=0.030$	181.1 $\pm$ 16.6	0.051			

Summary of one-way and two-way repeated measures ANOVA results for breath hold duration, spleen volume, and Hb concentration for each test condition before and after climbing Mount Everest. BH, breath hold; 1-W RM, one way repeated measures ANOVA; 2-W RM, two way repeated measures ANOVA. For within series effects p values <0.1 are denoted.

after three breath holds was not different during pre-and posttest, at 95.1 (1.5) % and 95.0 (1.4) % ($p=0.923$), respectively. Mean (SD) SaO_2 from exercise was not different during pre-and post-test, at 95.8 (1.1) % and 96.5 (0.77) % ($p=0.195$), respectively. During pretest mean (SD) $ETCO_2$ after exercise was 5.2 (0.4) %, and had increased by 5.8 % ($p=0.021$) from baseline. During post-test mean (SD) $ETCO_2$ after exercise was 4.5 (0.7) %, and did not differ from baseline ($p=0.126$). During pretest, mean (SD) RR after exercise was 25.4 (5.1) breaths/min, and had increased by 31.7 (16.2) % from baseline ($p=0.001$). During post-test, mean (SD) RR after exercise was 27.5 (2.7) breaths/min, and had increased by 33.4 (22.1) % from baseline ($p=0.005$). Compared to baseline, neither $ERCO_2$ ($p=0.535$) nor RR ($p=0.536$) were different during exercise in pre and post-test.

Spleen volume

Baseline values for spleen volume were similar between pre-and post-tests (Table 1). The mean spleen contraction was greater after the expedition (Fig. 1). A two-way repeated measure ANOVA revealed no time-condition interaction for spleen contraction. Separate consideration of pre and post spleen volume data (condition effect via one-way RM ANOVA) at BH1, BH2, BH3, exercise, and recovery revealed that the mean levels of spleen volume was not different from baseline in pre-altitude measurements, while it decreased from baseline in the post-altitude measurements. For post-altitude measurements, mean (SD) spleen volume during BH3 and exercise was reduced from baseline by 38.9 (17.2) % and 45.6 (15.1) %, respectively. After recovery, mean (SD) spleen volume was still reduced by 22.3 (13.5) % during post altitude measurements (Fig. 1).

Long-term hematological response

Baseline values for Hb were increased by 23.1 (9.9) % from pretest to post-test (Table 1).

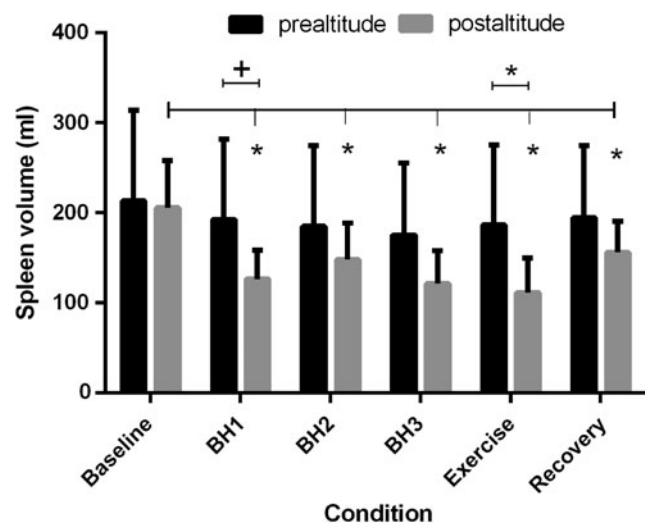


FIG. 1. Mean percent (SD) change in spleen volume from baseline immediately following each breath hold (BH1, BH2, BH3), exercise, and 10 min recovery before (pretest) and after (post-test) climbing Mount Everest. * denotes significance at $p<0.05$ in spleen volume reduction from baseline and between tests while + denotes trend at $p<0.1$ for difference of spleen volume between tests.

Hematological response to breath holding and exercise

The mean HB concentration was greater for post altitude measurements across all test conditions (Table 2). No significant time X condition interaction effect for Hb concentration was identified using two-way RM ANOVA. Separate analysis of pre and post Hb concentration data (condition effect via one-way RM ANOVA) revealed that the mean levels of Hb concentration tended to increase from baseline in the post-altitude measurements, while it was not different from baseline in the pre-altitude measurements (Table 2). Post hoc analysis of pre-altitude measurements showed that Hb concentration tended to be increased from baseline during exercise, while for post-altitude measurements it was increased from baseline during recovery (Table 2).

Discussion

This study shows that long-term high altitude exposure results in an increased spleen contraction both during breath holding and exercise, while resting spleen volume remains unchanged. This indicates that long-term hypoxic exposure may influence the contractile capacity of the spleen, with possible beneficial effects at altitude.

Both low PO_2 and high PCO_2 have been identified as potential triggers for spleen contraction (Lodin-Sundstrom and Schagatay, 2010; Richardson and others, 2009; 2012). In this study, baseline SaO_2 was similar in pre- and post-tests. Therefore, the chemical influence arising from O_2 may not explain the difference in spleen contraction. Although Richardson and others (2012) found that splenic contraction during breath holding increased following pre-apneic inspiration of a 5% CO_2 gas mixture compared to when air was used, it is not known whether the lower $ETCO_2$ at baseline during post-test might have affected spleen contraction after long-term altitude exposure.

Furthermore, the similar HR during exercise suggests that the level of cardiovascular stress were not an explanatory factor for the difference in spleen contraction.

This study suggests that prolonged exposure to high altitude may augment spleen contraction.

The augmented spleen contraction observed after the climb could be related to the long period of hypoxic and hypocapnic exposure typical of such a climb. Although not measured during the expedition, ascents of Mount Everest are characterized by long periods of systemic hypoxia typically interrupted by self-administered oxygen breathing at the greatest heights (West, 1983). Such climbs are also characterized by prolonged strenuous physical activity, which may be an additional explanatory factor for alterations in splenic contraction, as this would likely require that all means of increasing oxygen stores, including Hb concentration elevation, are recruited. The reduction of body mass and waist circumference indicate long-term catabolic metabolism, but it is not known if or how these factors may affect spleen contraction; in fact, little is known about the effects of prolonged high altitude exposure on human organ size. However, Sonmez and colleagues (2007) showed a reduction in spleen volume in lowlanders after 3–6 months relocation to high altitude, contrary to the current results where baseline spleen volumes remained unchanged. The difference in the duration and level of hypoxia exposure between the two studies could possibly explain the diverging results, or per-

haps that reference values in the Sonmez and colleagues (2007) study were not standardized for the hypoxic situation.

The further reduction of spleen volume during post-test tended to be accompanied by a greater increase in Hb concentration from baseline after breath holding and exercise tests. Lack of significant results could be due to either the small sample size, or possibly to a different peak time of Hb concentration after spleen contraction after altitude exposure. There is a time lag after spleen contraction to maximal elevation of Hb concentration, as the erythrocyte concentrate must have time to mix with the circulating blood (Schagatay and others, 2005). In seals, the spleen contents are, after contraction, initially stored in specialized venous pools and successively let into the systemic circulation (Fahlman, 2012). Even though such special features may be absent in man, a small delay when erythrocytes are released has been shown (Schagatay and others, 2005), and could possibly be affected by altitude exposure. Furthermore, the splenic contribution to breath hold-induced elevation of Hb concentration has previously been estimated to be approximately 60% (Richardson, 2008). Thus, other vascular areas may also contribute to this effect, which could possibly provide an explanation for the relative little enhancement of the Hb concentration elevation despite a greater spleen contraction.

In this study, a general polycythemia was evident after altitude exposure in line with previous studies (West, 2012). It is not known whether greater blood viscosity could affect splenic emptying. Since the more pronounced spleen contraction was not accompanied by a more substantial increase in Hb concentration, the functional aspects of the more pronounced spleen contraction is not clear. However, while both Hb concentration and the spleen volume returned to baseline values after 10 minutes of recovery during pre-test, Hb concentration still remained elevated and the spleen remained reduced after 10 minutes of recovery during post-test. This suggests that blood oxygen-carrying capacity remains elevated after long-term altitude exposure and thus suggests a greater efficiency of this response after altitude acclimatization.

The augmented spleen contraction after long-term hypoxic exposure is in line with the augmented spleen contraction and increased rise in Hb concentration previously observed in trained breath-hold divers (Richardson and others, 2005; Prommer and others, 2007; Schagatay 2010) who are also often exposed to hypoxia, and further suggests that these effects may be due to training and not only to genetic predisposition. Periodically-occurring hypoxia as achieved during breath holding may be a powerful stimulus potentially affecting the spleen's contractility in a similar manner as long-term hypoxic exposure. However, it should be noted that during breath holding $Paco_2$ is gradually increased, potentially giving rise to an enhanced spleen contraction (Richardson and others, 2012). However, during altitude ascent, respiration is increased and $Paco_2$ is decreased, which may involve decreased capnic influence on the spleen.

Limitations of the study

The volunteers were for practical reasons part of the same expedition aiming for the top of Mount Everest, and included both Sherpa and western low-landers with different altitude histories. It might be that the Sherpas differ in their splenic

responses to hypoxia due to an inborn genetic adaptation to high altitude. Furthermore, the Sherpas hypoxic exposure in Kathmandu before the expedition, and the many trips with great loads at high altitude during the climb, could possibly have affected their splenic contractile function, but due to the small sample size difference between ethnic groups could not be evaluated. A direction for further studies would be to compare the spleen's contractile function in ethnic lowlanders and highlanders.

Conclusions

We concluded that long-term altitude exposure augments and prolongs spleen contraction in response to acute hypoxic challenges, while baseline spleen volume remains unchanged. A more efficient spleen contraction could be beneficial by elevating Hb concentration during work at high altitude, while storing erythrocytes temporarily not needed during rest in the spleen would reduce the work of the heart caused by the polycythemia.

Acknowledgments

We thank the climbers for participating in our study despite their tight schedule, the expedition leader Mr. Chhiring Dorje Sherpa for the opportunity to render the study, him and his co-workers Mr. Ngawang Tashi Sherpa and Mr Mingma Tenzing Sherpa at Rolwaling Excursions for much help during our visits in Nepal, and Mr. Stefan Jakobsson for caring for our laboratory between measurements. We gratefully acknowledge the advice of Dr. Matt X. Richardson who worked in the field in this region before.

Author Disclosure Statement

No competing financial interest exists. This study was supported by the Swedish National Centre for Research in Sports (CIF) and the Swedish Winter Sports Research Centre at Mid Sweden University.

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